

A Deep Learning Pipeline for Human Activity Recognition

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Outline

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Goals for the project	3
Data collection	4
Preprocessing & Segmentation - IMU data	6
DL Architecture	9
Max-Pooling vs. Absmax-Pooling	10
“Small” model results - 6-Fold Cross-Validation	11
Sitting vs. standing - “Big” model	12
Preprocessing - Pressure sensor signal	13
An unsuccessful attempt	14
State modelling	15
“Big” model results	16
Future work	16

Goals for the project

Goal

Collect data for HAR from 30+ participants

Implement an entire preprocessing pipeline

Finally understand how Git versioning works

Realize Pytorch is exhausting but infinitely more flexible

Build custom pooling layers from scratch to fit domain

Build a DL model that accurately classifies activities

Utilize sensor fusion to differentiate sitting v. standing

Workout dataset analysis & calorie estimation

Gait analysis

Have fun 😁

Achieved?

✓✓ (42!!!)

✓

✓ 🤔

👻👻

✓ (somewhat novel!)

✓✓

📈📈

Ana's BSc Thesis <3

Tomi's BSc Thesis <3

✓✓✓

Data collection

- ▶ Collect data from **42** participants in InnoFEIT and Emteq
- ▶ Supervise and annotate data for each participant individually
- ▶ Collected data for 5 tasks:
 - Everyday activities (our focus)
 - Reading
 - Transitions (standing up, sitting down, laying down...)
 - Workout (Ana's future work)
 - Gait (Tomi's future work)

Data collection

Imagine a month and a half of this...

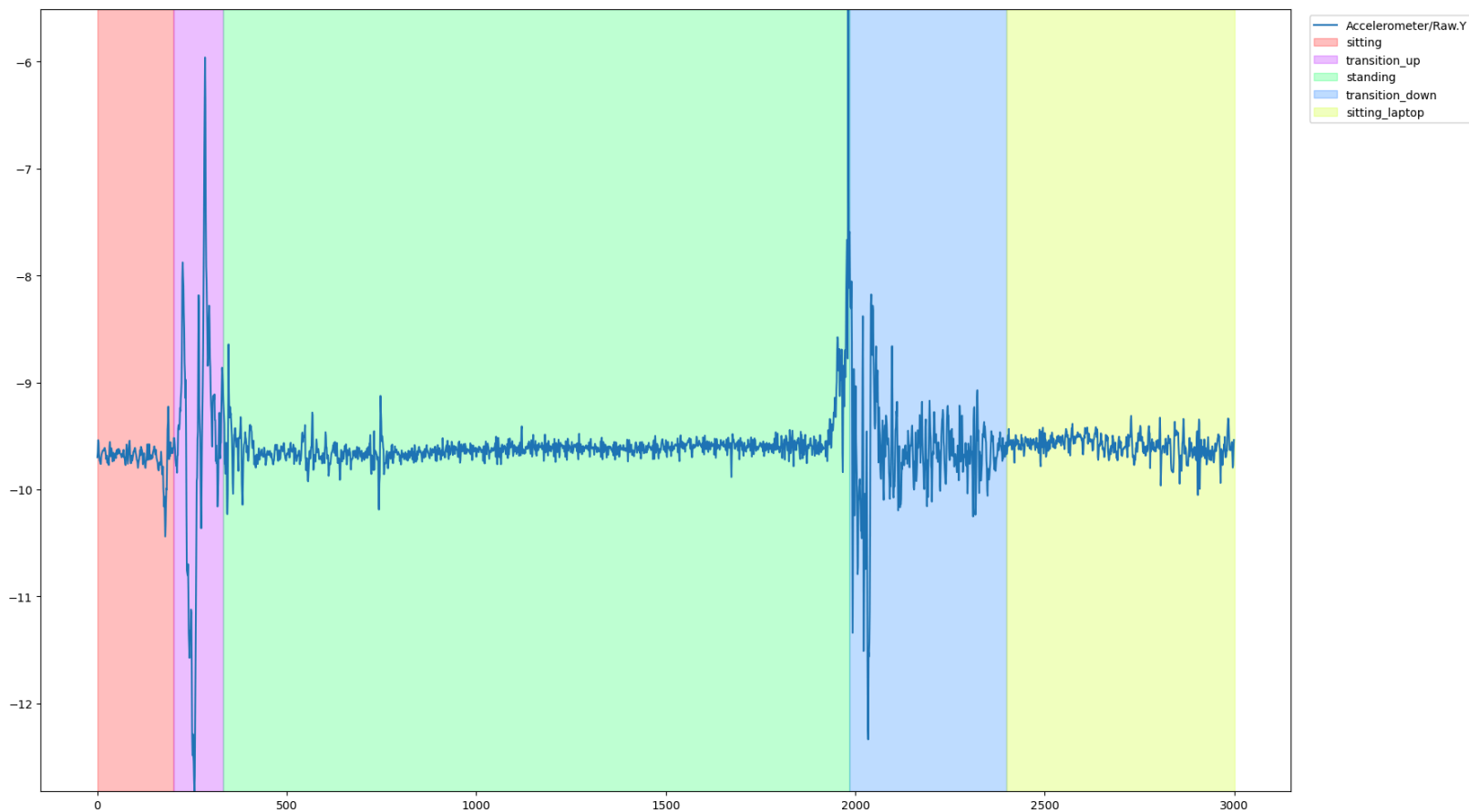


Preprocessing & Segmentation - IMU data

- ▶ Accelerometer (X, Y, Z) ✓
- ▶ Gyroscope (X, Y, Z) ✓
- ▶ Magnetometer (X, Y, Z) ✗ (too sensitive to external conditions)
- ▶ Outlier isolation & smoothing - via rolling window average
- ▶ Low-Pass Butterworth filter (5th order) - $f_{\text{high_cutoff}} = 4$ [Hz]
 - Attenuates high-frequency sensor noise, preserves low-frequency structural components of body movement.
- ▶ Euclidean norms: $A_{\text{mag}} = \sqrt{A_x^2 + A_y^2 + A_z^2}$, $G_{\text{mag}} = \sqrt{G_x^2 + G_y^2 + G_z^2}$
 - Same preprocessing applied as to individual channels
- ▶ $W = 4[\text{s}] = 200$ samples, $S = 1[\text{s}] = 50$ samples; 75% overlap

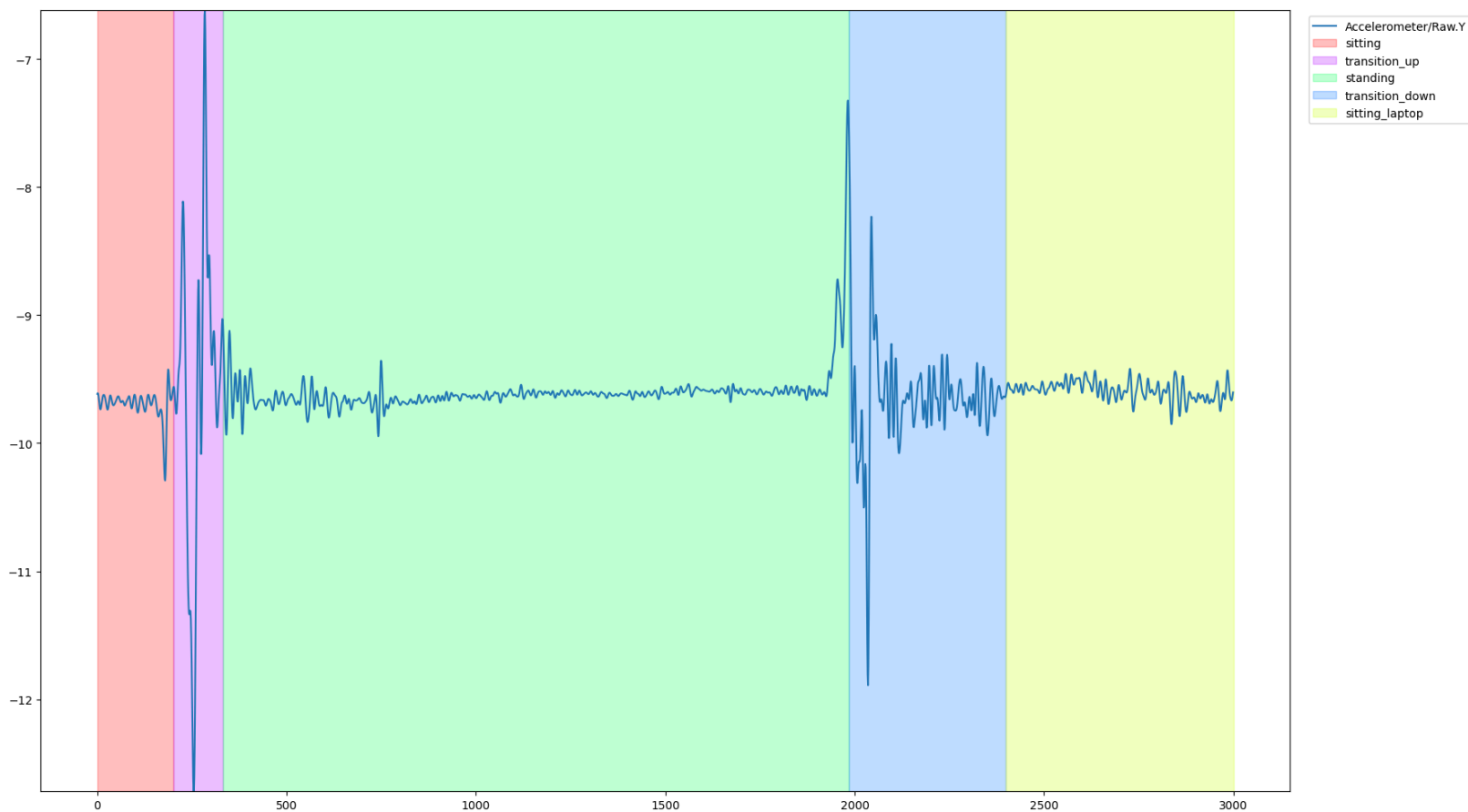
Effect of filtering

► Before filtering



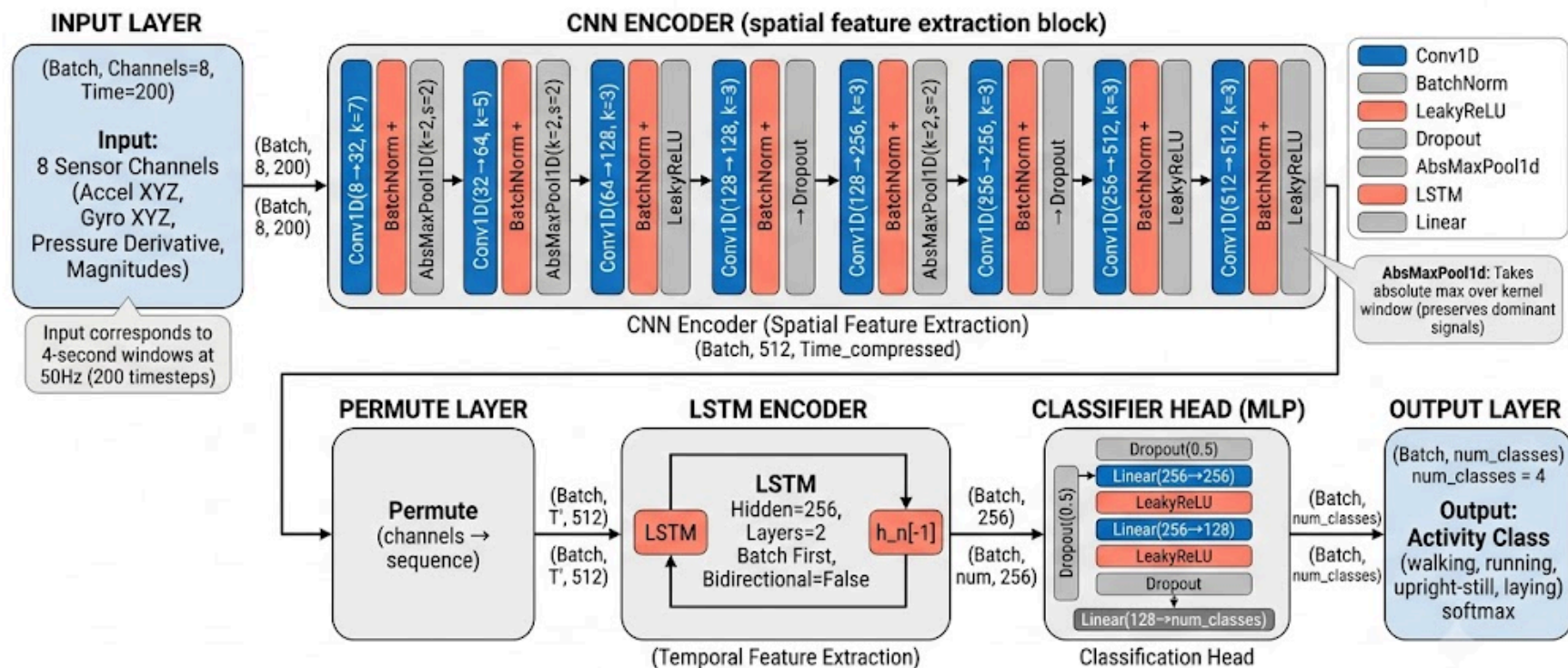
Effect of filtering

► After filtering



DL Architecture

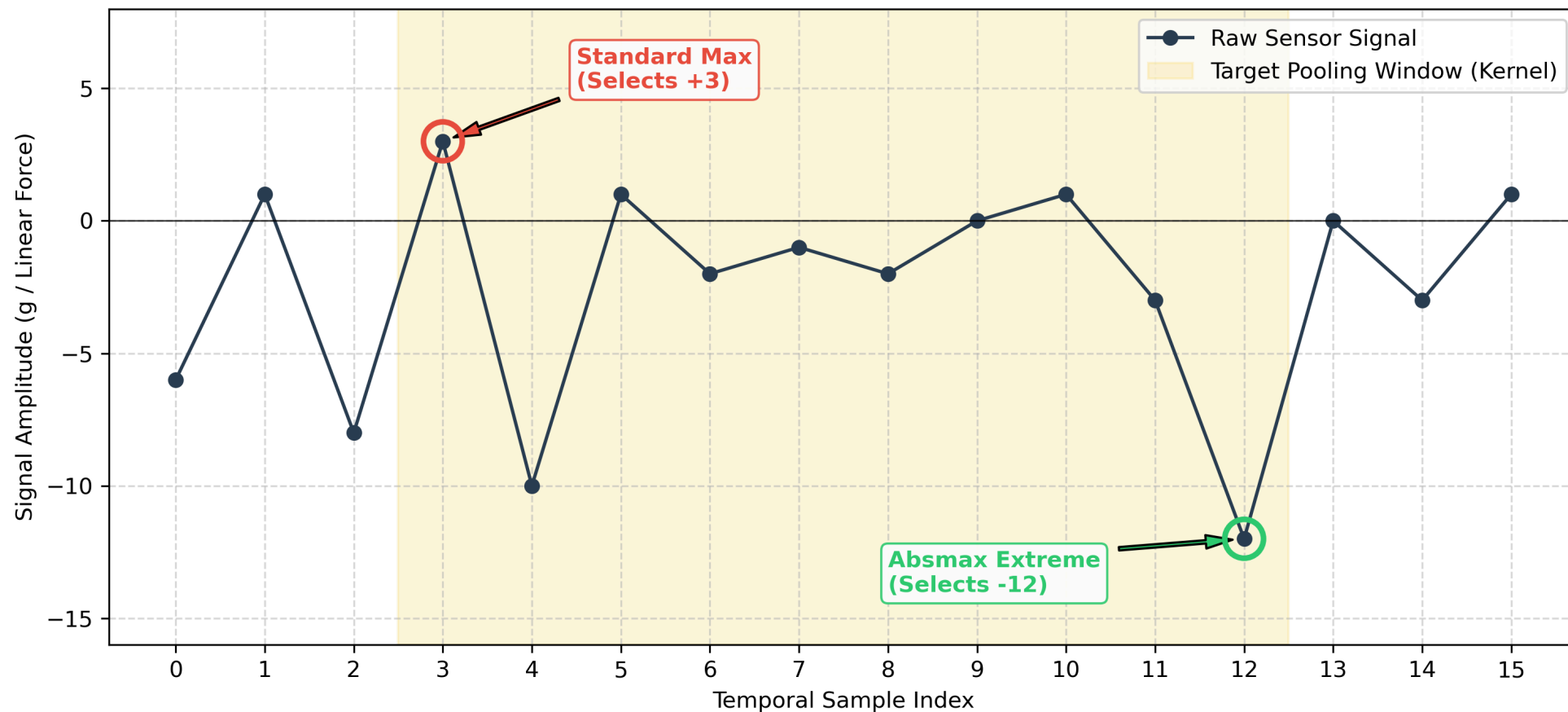
CNNLSTMAbsMax Architecture for HAR



Total trainable parameters:	[visible as CNNEncoder + LSTM + ClassifierHead]
Input:	8 channels × 200 timesteps
Sensor columns:	Accelerometer (X,Y,Z,Magnitude), Gyroscope (X,Y,Z,Magnitude), Pressure Derivative

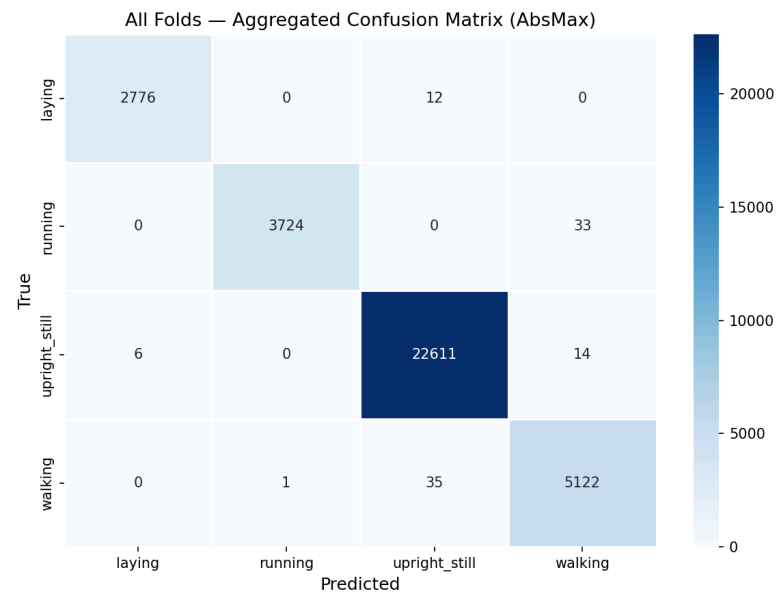
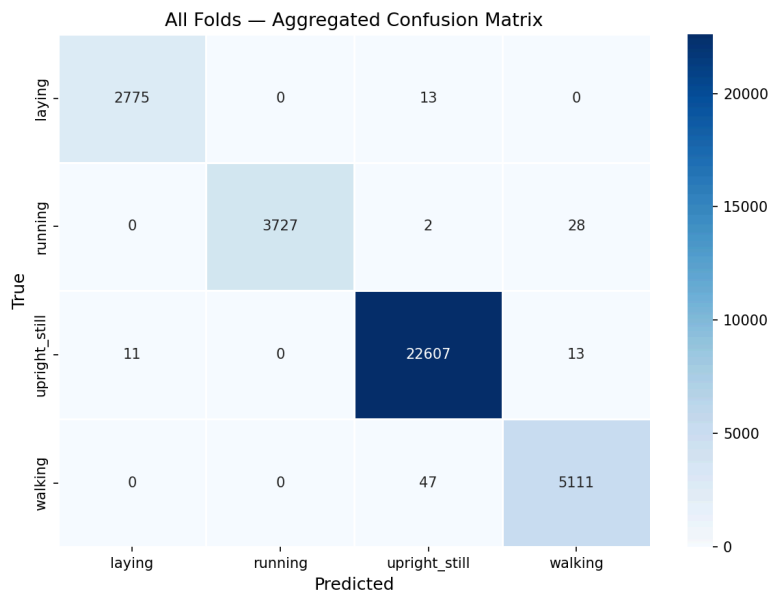
Max-Pooling vs. Absmax-Pooling

Signal Sign Bias: Standard Max-Pooling vs. Absmax-Pooling



“Small” model results - 6-Fold Cross-Validation

Table:



Sitting vs. standing - “Big” model

Preprocessing - Pressure sensor signal

An unsuccessful attempt

State modelling

“Big” model results

Q&A

Thanks <3